Speech and Audio Processing

03 - speech analysis and deep learning

Shinnosuke Takamichi / 高道 慎之介

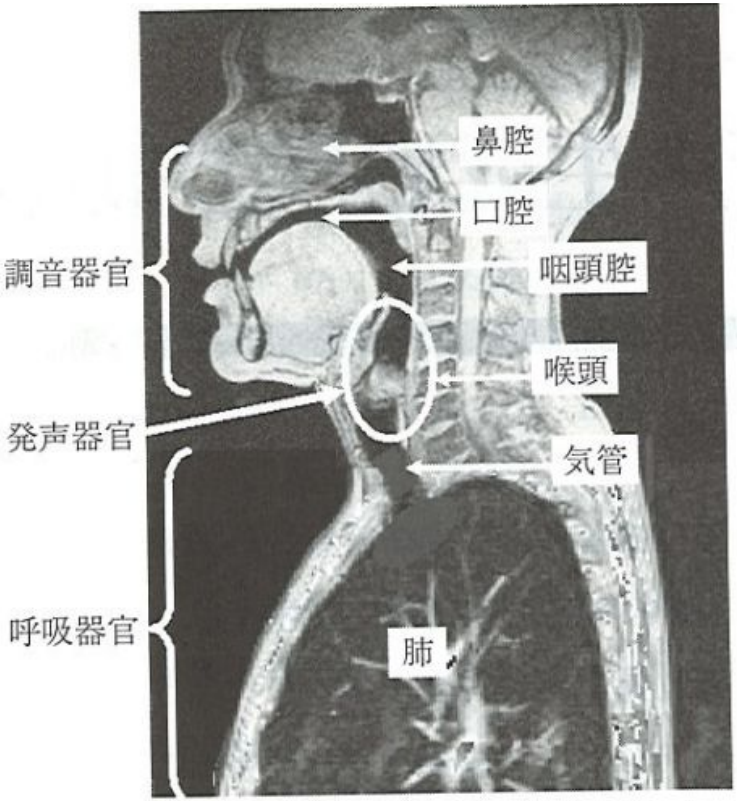
Takamichi Laboratory
慶應義塾大学 高道研究室

Class schedule (tentative)

01	Oct. 02	Introduction
02	Oct. 09	Speech production / 音声生成
03	**Oct. 16**	**Speech analysis and deep learning / 音声分析と深層学習**
04	Oct. 23	Speech recognition / 音声認識
05	Oct. 30	Self-supervised learning / 自己教師あり学習
06	Nov. 6	**Guest Talk (acoustic signal processing) / ゲストトーク (音響信号処理)**
07	Nov. 13	Foundation model / 基盤モデル
08	Nov. 27	Advanced topics / 発展的話題
09	Dec. 04	**Guest Talk (music information processing) / ゲストトーク (音楽情報処理)**
10	Dec. 11	Speech system building / 音システム構築
11	Dec. 18	Speech system building / 音システム構築
12	Dec. 25	Speech system building / 音システム構築
13	Jan. 08	Speech system building / 音システム構築
14	Jan. 22	Final presentation / 最終発表

Last week's class

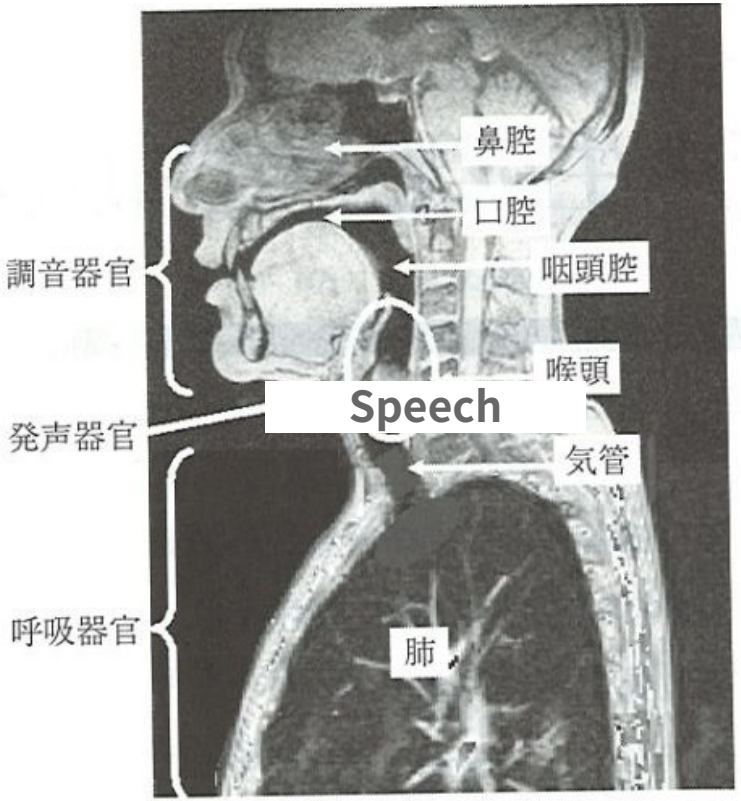
What happens in our body when producing speech





Today's class

How is feature extracted? How are neural networks used to model?



Analysis

Synthesis

Feature

Recognition

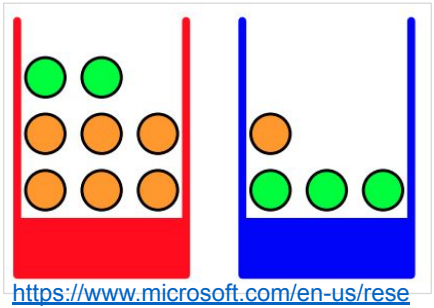
Synthesis

Label

Deep learning / 深層学習

Let’s review mathematics and fundamentals of deep learning!

Probability / 確率



https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf

Q1. A ball was drawn at random 10 times. When choosing the red box 4 times and blue box 6 times, what is the **probability of choosing each box**?

$$P(x = \text{red}) = \frac{4}{10} \qquad P(x = \text{blue}) = \frac{6}{10}$$

Sums to 1

(Discrete) random variable

Always non-negative

Q2. A ball was drawn at random once. **Given that the chosen box was blue, find the probability of picking an orange ball. (= Conditional probability)**

$$P(y = \text{orange}|x = \text{blue}) = \frac{1}{4}$$

Event that already happened (= Condition)

Q3. A ball was drawn at random once. **Find the probability that the chosen box is blue AND an orange ball is picked. (= Chain rule)**

$$P(y = \text{orange}, x = \text{blue}) = P(y = \text{orange}|x = \text{blue})P(x = \text{blue})$$

Probability of x AND y

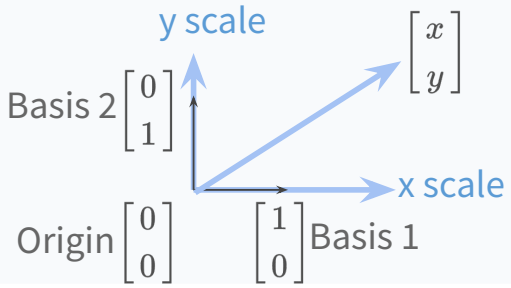
Probability of y given x

Probability of x

Linear algebra / 線形代数

Vectors represent how much basis vectors are scaled

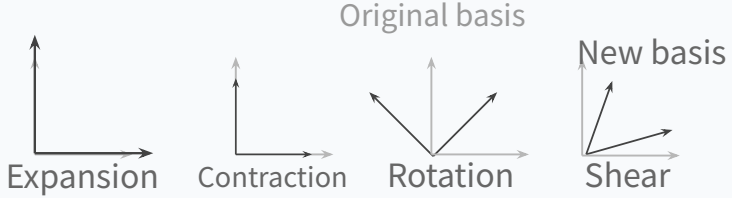
$$\begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Linear transformation swaps basis vectors. Matrix is a collection of new basis vectors

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$$

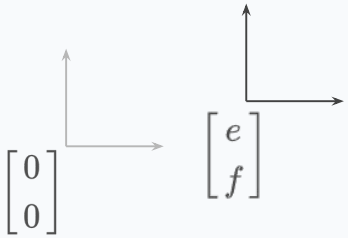
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + by \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}$$



Affine transformation swaps & shifts basis vectors

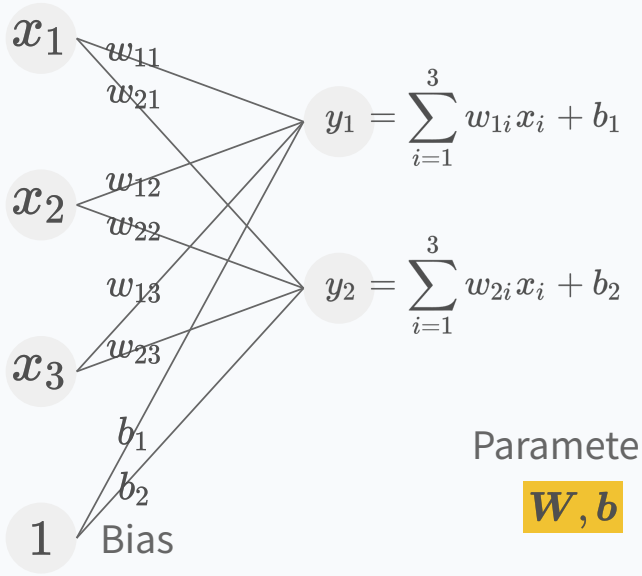
$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} ax + by + e \\ cx + by + f \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix}$$



Neural network: simple perceptron and activation

Single-layer perceptron without activation function is equivalent to affine trans.

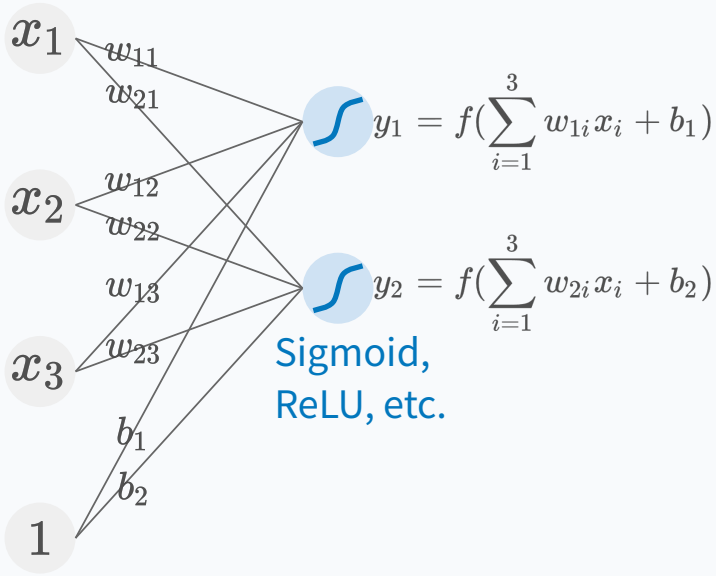


Parameters

$\boldsymbol{W}, \boldsymbol{b}$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \overset{\boldsymbol{W}}{\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \overset{\boldsymbol{b}}{\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}}$$

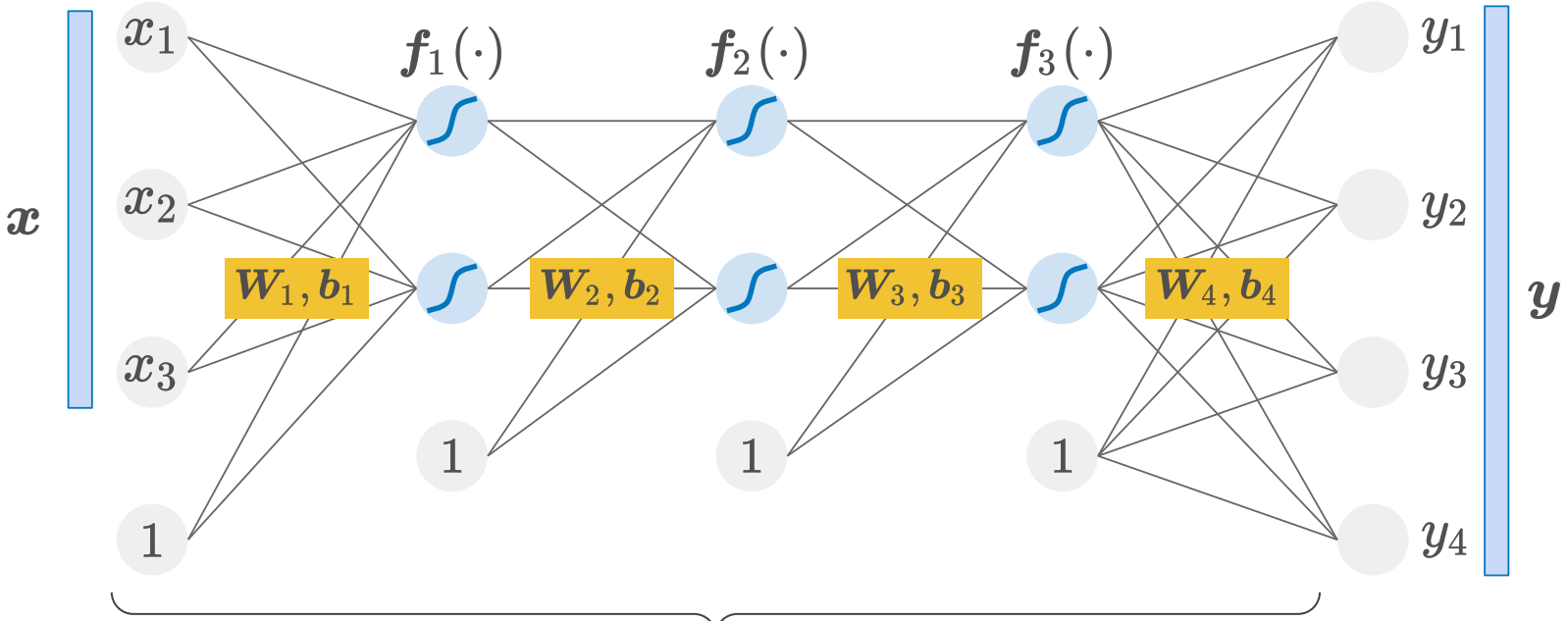
Activation functions enable non-linear transformation.



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = f\left(\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$$

Neural network: multilayer perceptron (MLP)

Stacking simple perceptrons enables "deep" non-linear transformations

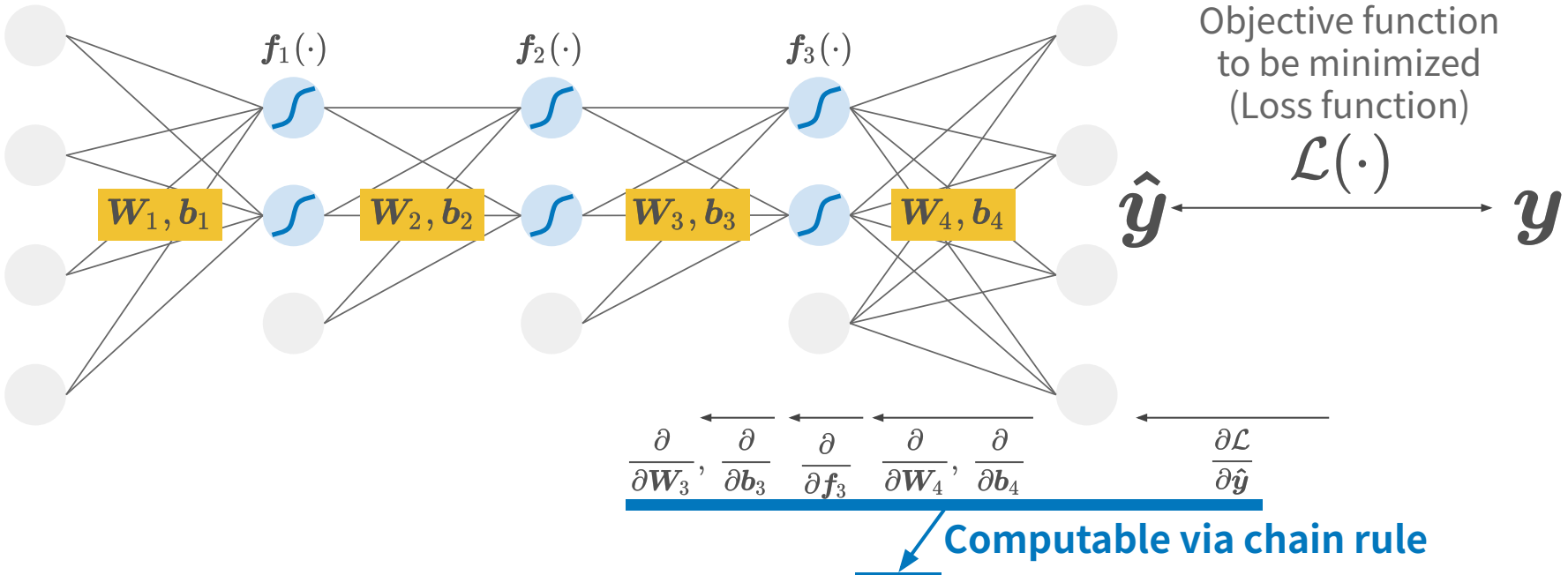


$$y = f_\theta(x)$$

Set of W_*, b_* (function with parameters θ)

Neural network: backpropagation

Parameters W, b can be learned via backpropagation.



$$W_3 \leftarrow W_3 - \alpha \frac{\partial \mathcal{L}}{\partial W_3}$$

After update Before update Negative gradient of objective function

Neural network model as statistical model (discrete probability distribution)

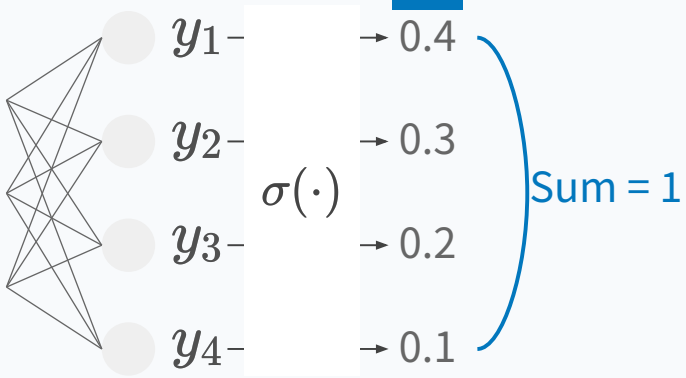
Softmax: propotion of exponentialized values

Proportion of the values

$$\sigma(\boldsymbol{y})_j = \frac{\exp(y_j)}{\sum_{\text{all } k} \exp(y_k)}$$

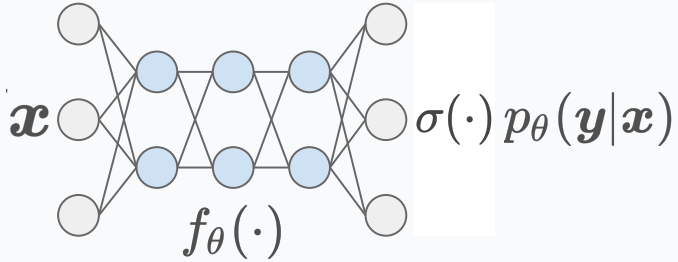
Sum of exponentialized values

Always non-negative

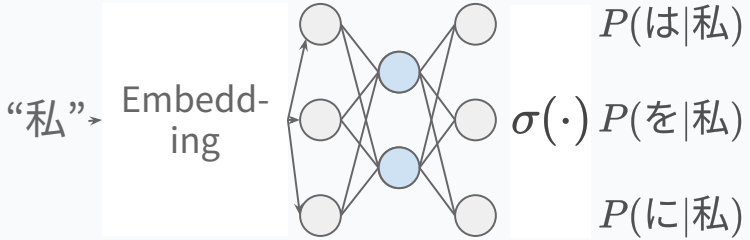


Outputs satisfy probability conditions.

Neural network that outputs probability values



Conditional probability of y given x, calculated with model parameters θ



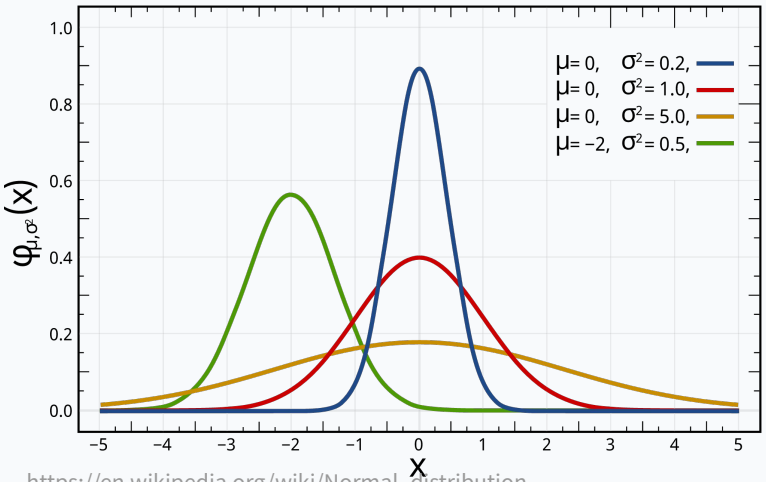
This neural network calculates conditional probability conditioned on a word

Neural network model as statistical model (continuous probability distribution)

Gaussian distribution (Normal distr.)

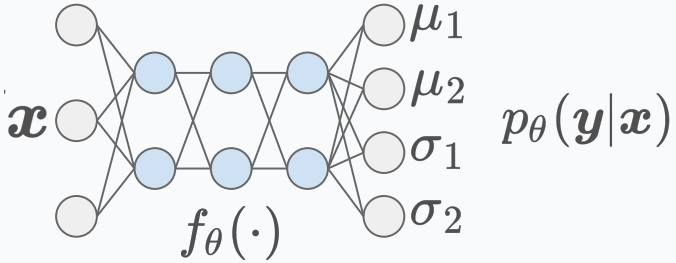
Mean μ, standard deviation σ.

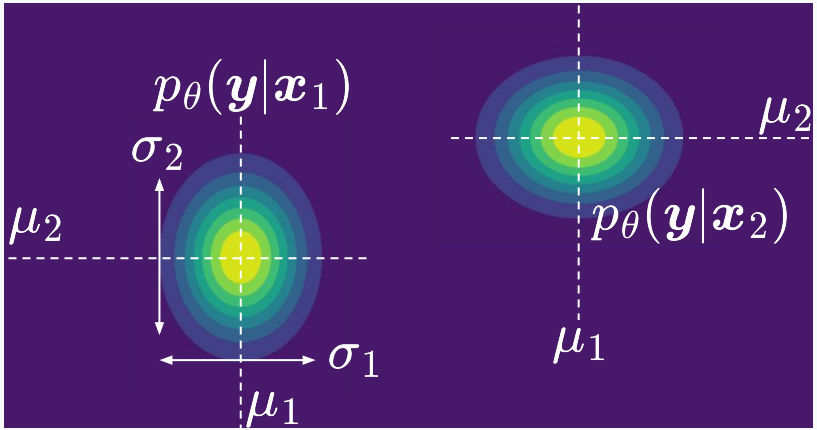
$$N\left(x;\mu,\sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



https://en.wikipedia.org/wiki/Normal_distribution

Neural network that outputs probability values





https://www.geeksforgeeks.org/python/visualizing-the-bivariate-gaussian-distribution-in-python/

Neural network model that transforms probability distribution

Variable transformation of probability distribution / 確率分布の変数変換

$$N\left(x;\mu,\sigma^2\right)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$x'=ax+b$$

$$N\left(x';\mu',\sigma'^2\right)=\frac{1}{\sqrt{2\pi\sigma'^2}}\exp\left(-\frac{(x-\mu')^2}{2\sigma'^2}\right)$$

$$\mu'=a\mu+b$$

$$\sigma'=|a|\sigma$$

Neural network that transforms prior probability distribution

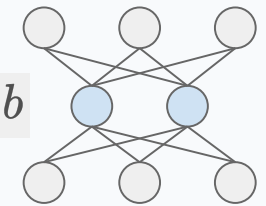
VAE, GAN, Flow, flow matching, etc.

$N\left(x;0,1\right)$

Samples data from the distribution.

If NN works as:

$f\left(x\right)=ax+b$



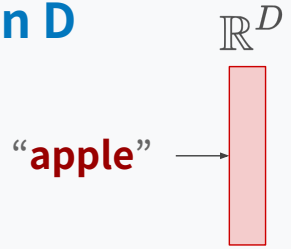
Output data follow the new distribution.

$N\left(x;b,|a|\right)$

Word embedding (token embedding)

Map to a dense vector (= embedding) of predefined dimension D

- Set of words (Vocabulary) : $V = \{w_1, \ldots, w_i, \ldots, w_{|V|}\}$
- Mapping : $f : V \to \mathbb{R}^D$



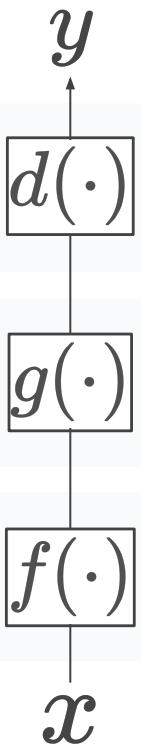
Mapping by word embedding matrix (lookup)

Extracting a specific row of a matrix. The so-called Word Embedding layer.

$$\underbrace{\begin{bmatrix} -0.1 \\ 3.0 \end{bmatrix}}_{\mathbb{R}^D} = \underbrace{\begin{bmatrix} -0.1 & 0.2 & 0.3 \\ 3.0 & 2.0 & 1.0 \end{bmatrix}}_{\mathbb{R}^{D \times |V|}} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{one-hot}}$$

"apple"

Prediction of y from x



Decision

- Selecting one option from among output candidates

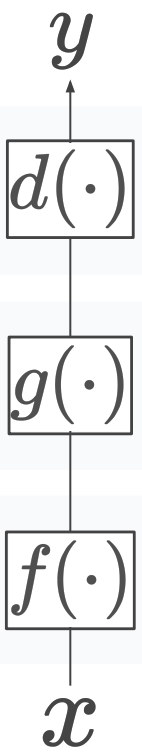
Score calculation

- Calculating output candidates and their scores based on features

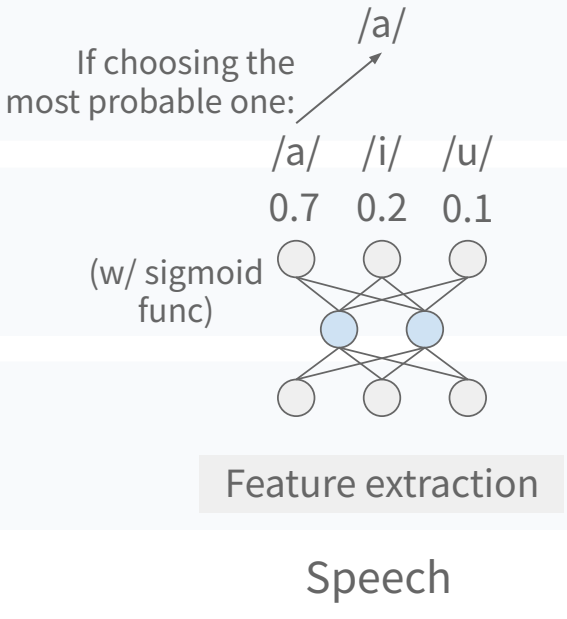
Feature extraction

- Extracting meaningful features from input

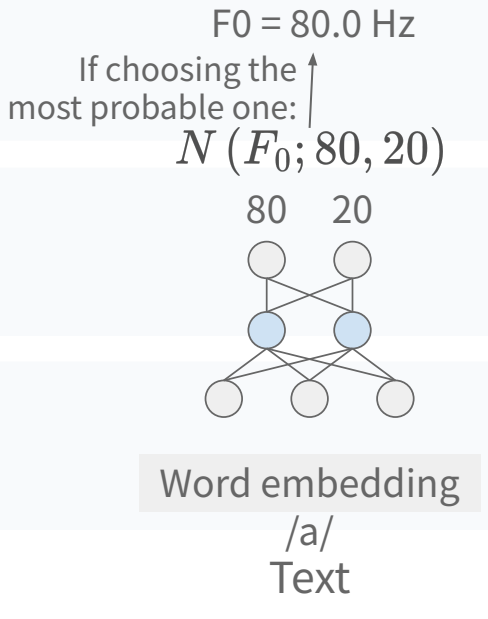
Example



Speech-to-text (cont. to disr.)



Text-to-speech (disr. to cont.)



Speech analysis / 音声分析

Revisiting speech production



Let's take a look at the speech waveform.

1. Find periodicity (sometimes aperiodicity) in the local frame.
2. What happens when **pitch** is changed.
    a. The period = period of open-close cycle = inverse of F0
3. What happens when **phoneme** is changed.
    a. The one-period waveform = phoneme
4. What happens when **speaker** is changed.

(Please try later by yourself, using audacity.)

Speech features from local to global



< 5~10msec: spectrum, F0

< 50~150msec: phoneme

< a few sec: accent, intonation

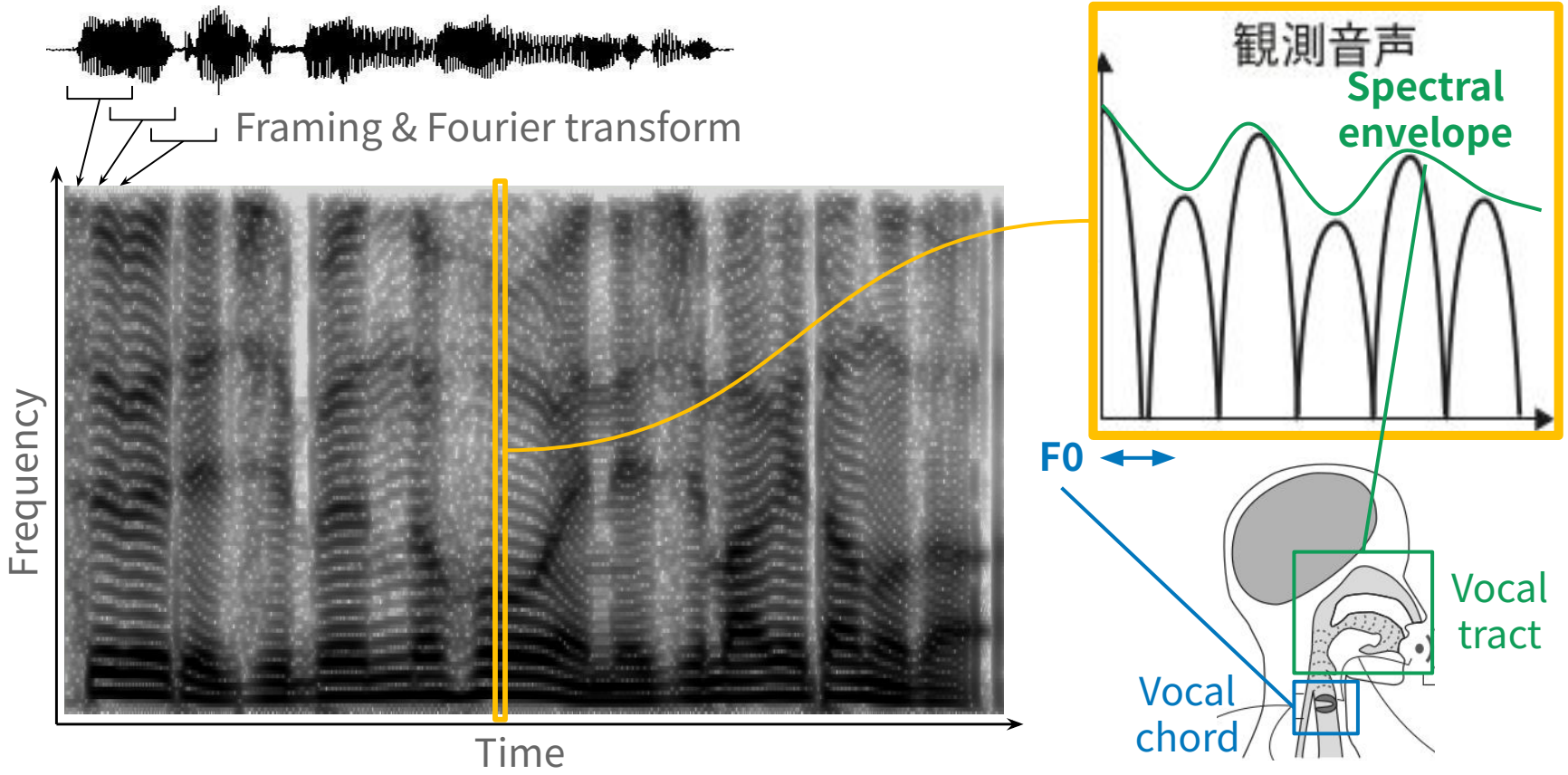
E.g., rising pitch -> question,
falling pitch -> confirmation

Const.: speaker's attribute
(identity, gender, age, dialect, etc.)

Local

Global

Frame analysis and spectrogram



Extraction of spectral envelope and F0

- Spectral envelope:
  - Traditional signal processing:
    - Linear predictive coding (LPC): assuming resonance in vocal tract
    - Cepstrum analysis: based on Fourier transform
  - Recent signal processing: WORLD vocoder

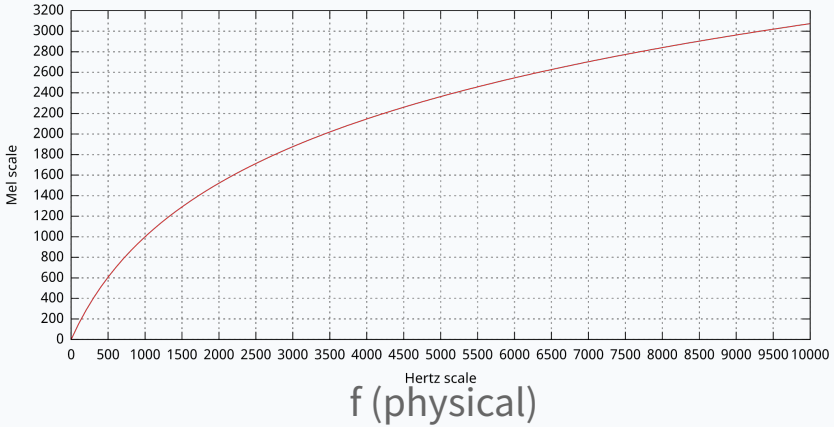
- F0:
  - Traditional: zero-cross, cepstrum, etc
  - Recent: WORLD (dio), RAPT, SWIPE, pYIN, CREPE, etc.
  - Deep learning: FCNF0++, YOLOPitch, etc.

Mel scale

Mel scale / メル尺度

Human's perceptual scale of pitches (human's perception is not linear to pitch increase.)

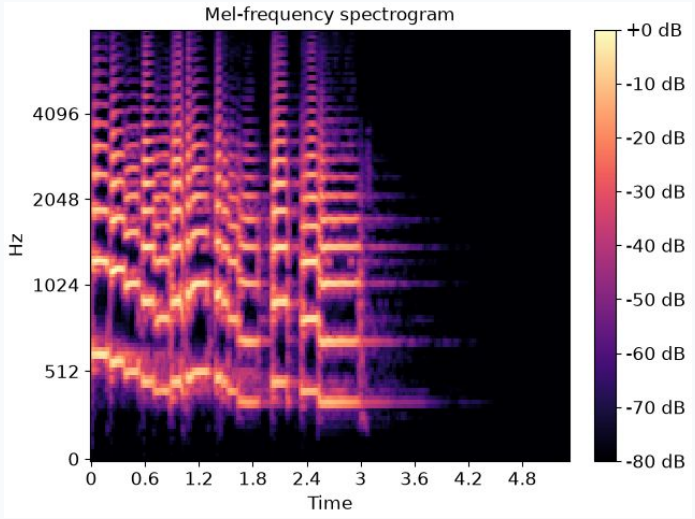
$$\mathrm{mel} = \frac{1000}{\log_{10} 2} \log_{10}\left(\frac{f}{1000} + 1\right)$$



https://en.wikipedia.org/wiki/Mel_scale

Mel-spectrogram / メルスペクトログラム

Transforming freq axis into mel-scale.
of freq bins can be reduced from linear-frequency spectrogram.

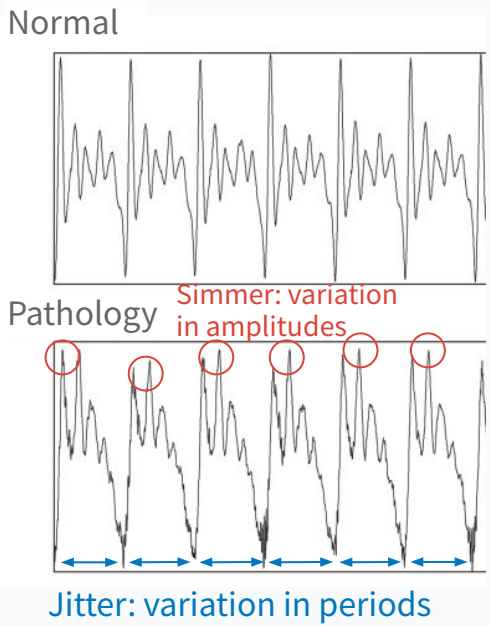


https://librosa.org/doc/0.11.0/generated/librosa.feature.melspectrogram.html

Other (local) features

Jitter & Simmer / ジッター, シマー

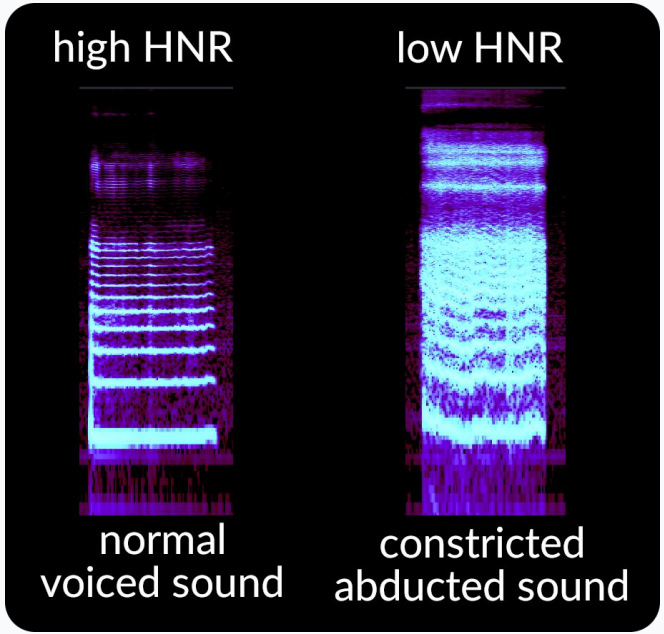
Variation in/of one period



https://www.jstage.jst.go.jp/article/larynx/28/2/28_78/_pdf

HNR (harmonic-noise ratio)

Ratio of harmonics and non-harmonics, breathiness (high HNR), creaky (low HNR)



Vocoder / ボコーダ

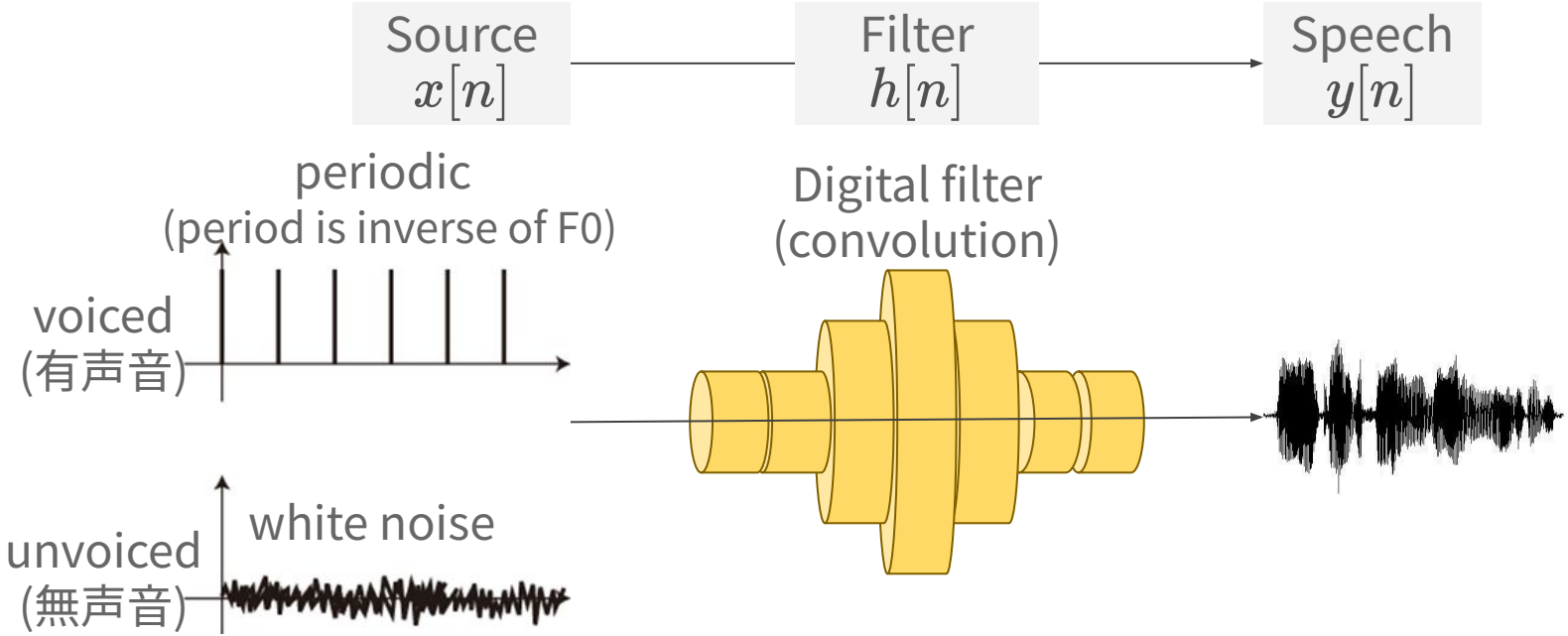
- Vocoder = voice + coder
  - Encoder: parameterize features from speech waveform
  - Decoder: synthesize speech waveform from features



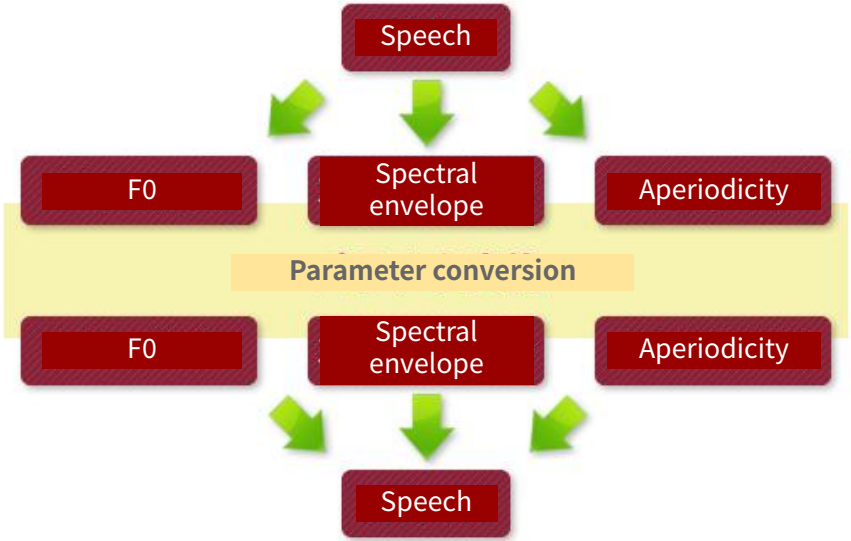
- Types
  - Phase vocoder (フェーズボコーダ): based on amplitude and instantaneous phase
  - Sinusoidal vocoder (正弦波ボコーダ): additive sine wavforms
  - Channel vocoder (チャネルボコーダ): based on source-filter model
  - Neural vocoder (ニューラルボコーダ): based on neural networks

Signal processing-based vocoder

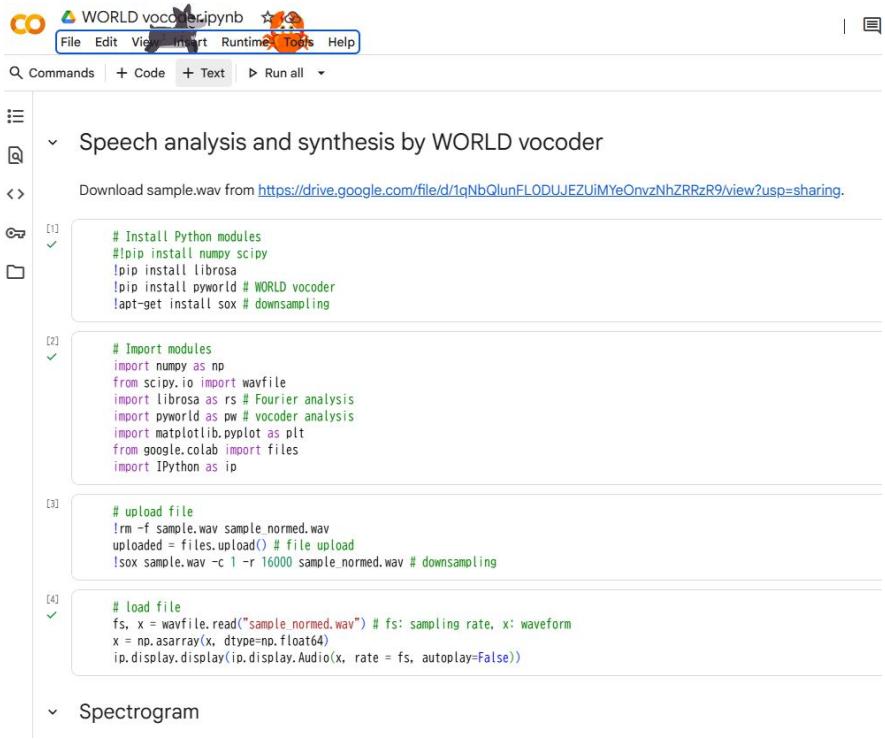
Based on the source-filter theory, source and filter components are parameterized.



WORLD vocoder



https://www.isc.meiji.ac.jp/~mmorise/world/introductions.html



https://colab.research.google.com/drive/1FS2Wa28ISuXxY_cvFvcHuuiSFv9v0CiH?usp=sharing (please copy to your gdrive.)

Speech analysis using deep learning / 深層学習を用いた音声分析

Vocoder is one kind of speech (audio) codec



- **Interpretability:** Corresponds to the speech production and information.
- **Controllability:** Elements can be controlled independently and intuitively.
- **Compression:** Can be represented with a small number of parameters.
- **Computational complexity:** Fast & lightweight encoding / decoding.
- **Audio quality:** High quality of decoded audio.

Speech Codec and Vocoders



Signal processing

Speech/audio codec

PCM/ADPCM
MP3/MPEG
AAC/EVS
…

CELP
ACELP/VCELP
…

Vocoder

LPC/PARCOR
HNM/Sinusoidal
STRAIGHT/WORLD
…

Machine learning

Neural speech/audio codec

SoundStream
Encodec
SoundStorm
…

DAC
FAcodec
…

Neural vocoder

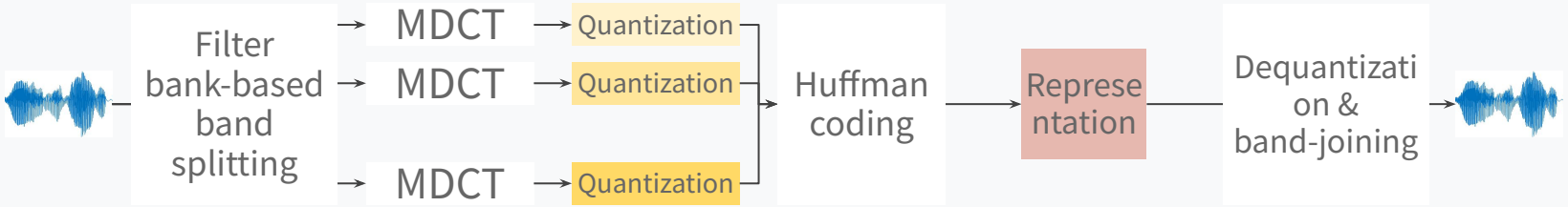
HiFi-GAN/BigVGAN
WaveGrad
WaveRNN
…

Audio codec
(for general audio)

Hybrid

Analysis-synthesis
(assuming speech)

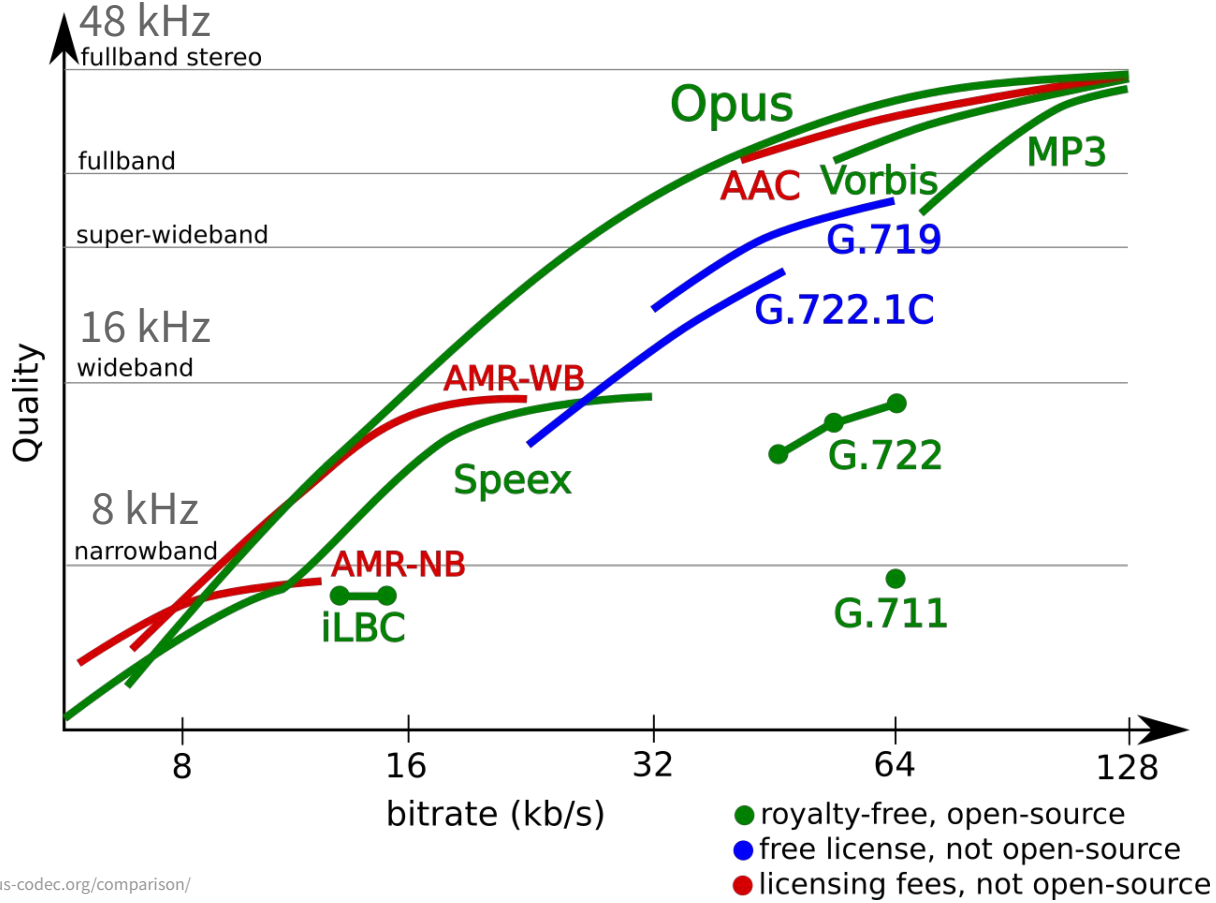
MP3: well-used audio codec



705.6 kbps (44.1 kHz x 16 bit)	320 kbps	192 kbps	128 kbps	64 kbps
Uncompressed audio (WAV)				

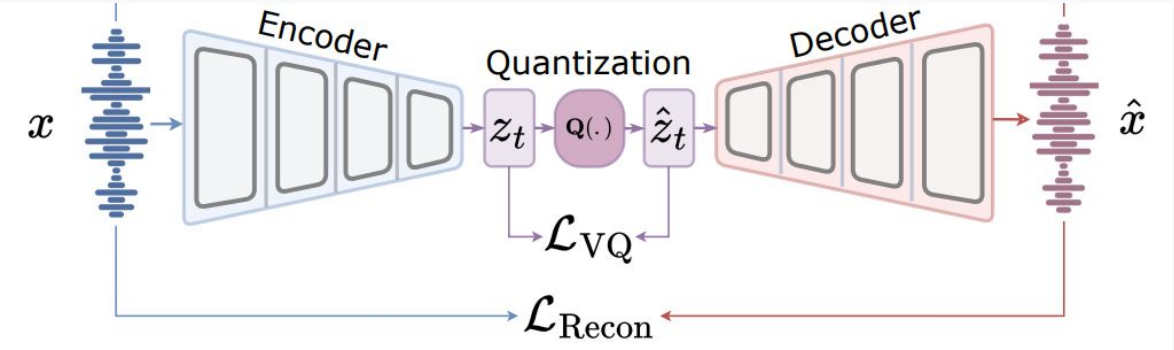
https://note.com/8nm/n/nca58667e8b83

Opus (based on signal processing) is more modern codec than MP3



https://opus-codec.org/comparison/

From signal processing to neural network



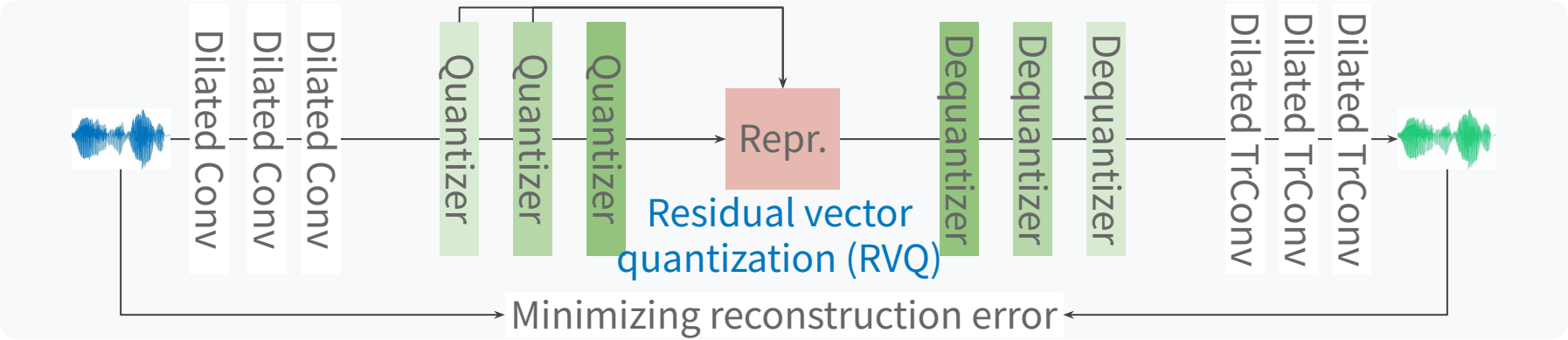
https://arxiv.org/pdf/2506.10274

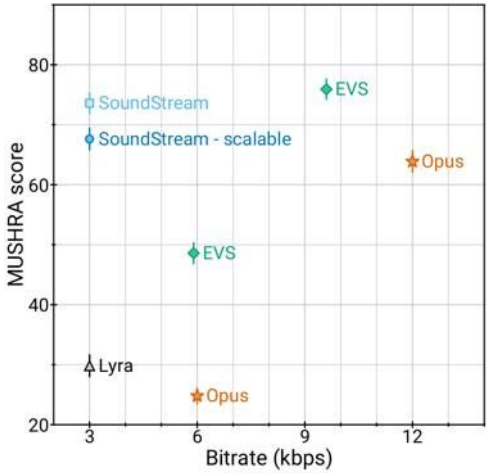
- **Data-driven:** Higher compression learned from big data
- **Post-refinement:** Decoder restores speech from lossy representation.
- **Superresolution:** low-freq range to high-freq range

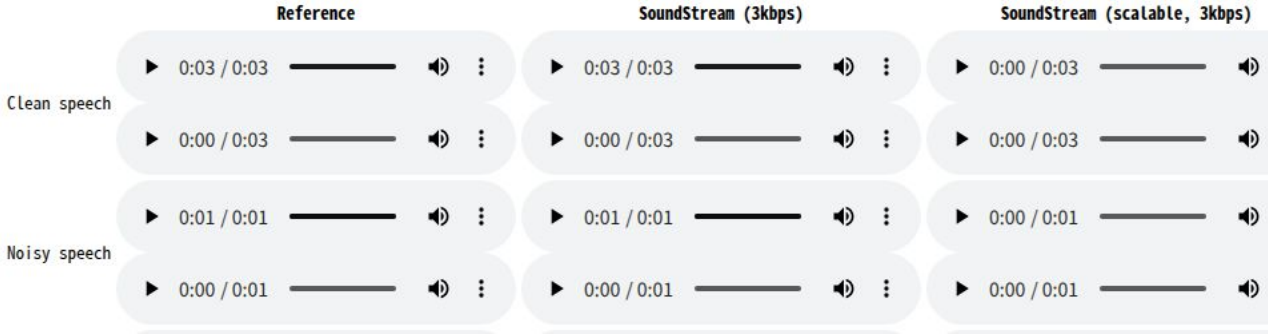
What's important to neural speech/audio codec.

- **Model:** Higher compression learned from big data
- **Quantization:** Decoder restores speech from lossy representation.
- **Training:** low-freq range to high-freq range
- **Domain:** low-freq range to high-freq range
- **Streamability:** low-freq range to high-freq range

SoundStream opened the door of neural codec.

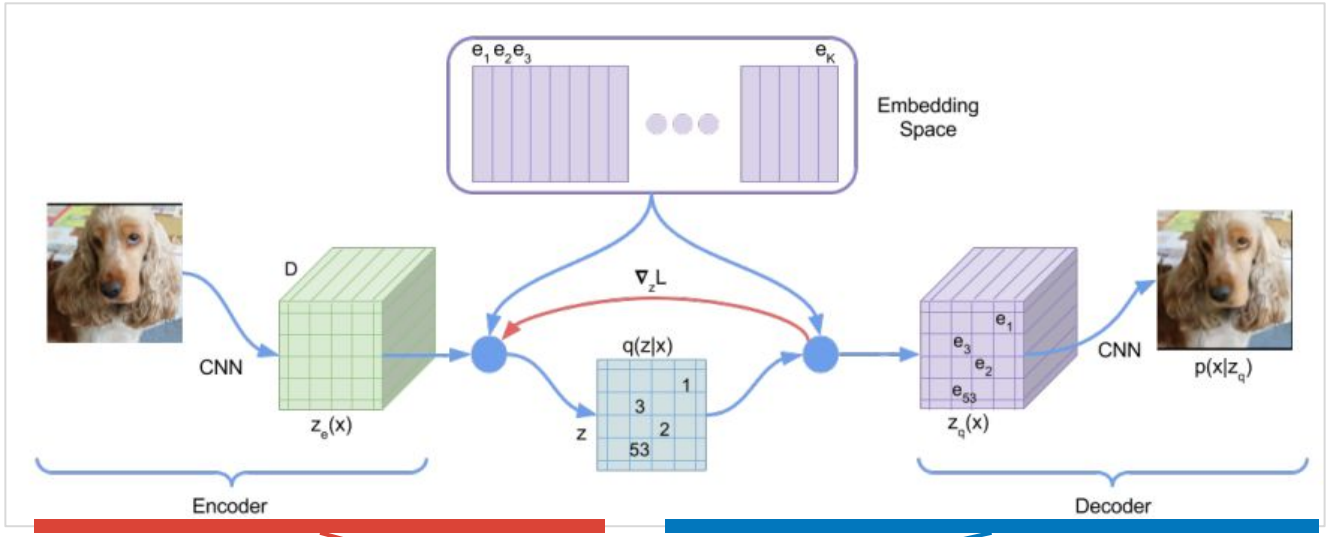






https://google-research.github.io/seanet/soundstream/examples/

VQ-VAE: variational auto-encoder to learn quantized representations

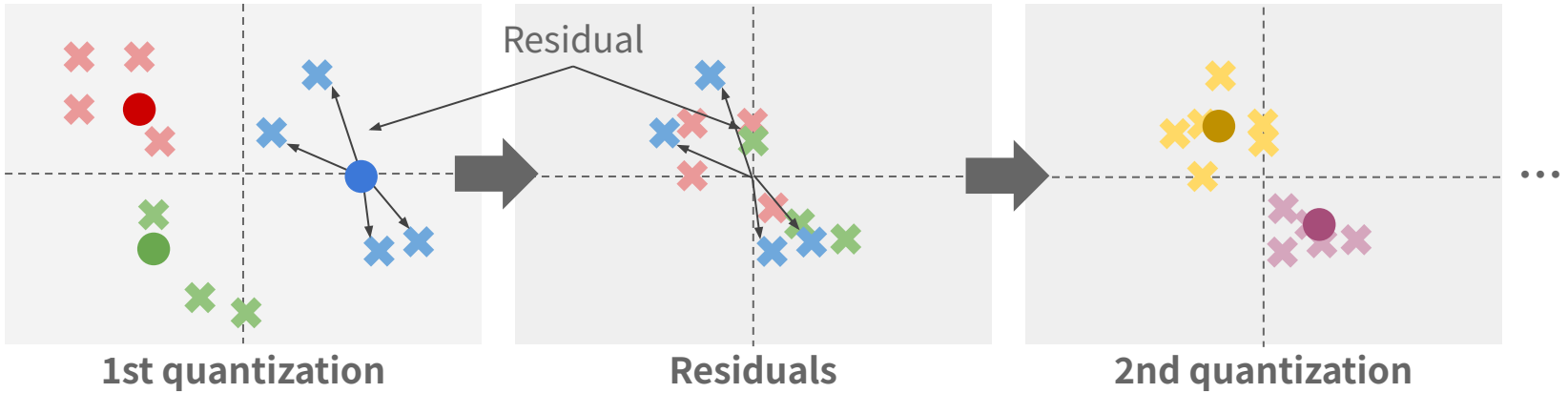


Learn decoder to reconstruct the input from the representations.

Learn encoder and quantized representations

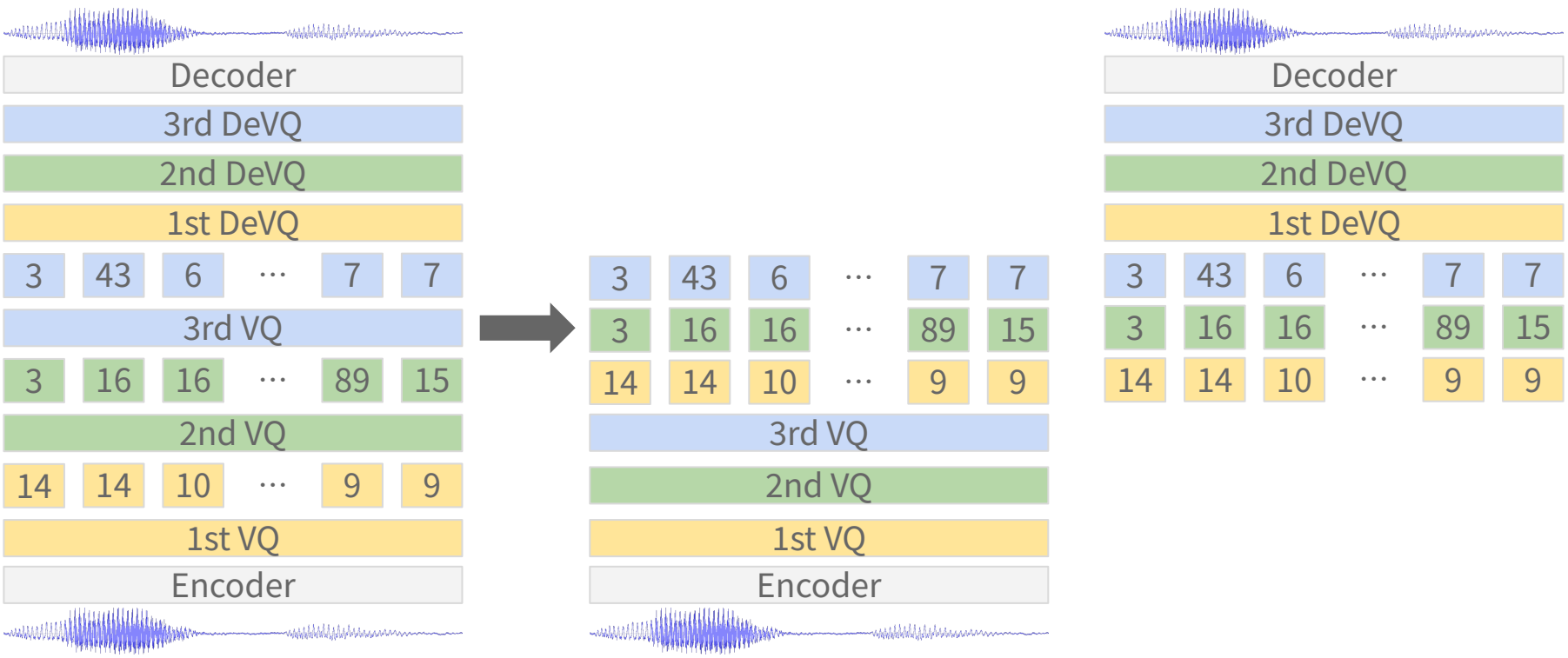
$$L = \log p(x|z_q(x)) + \|\text{sg}[z_e(x)] - e\|_2^2 + \beta\|z_e(x) - \text{sg}[e]\|_2^2,$$

Residual vector quantization (RVQ)



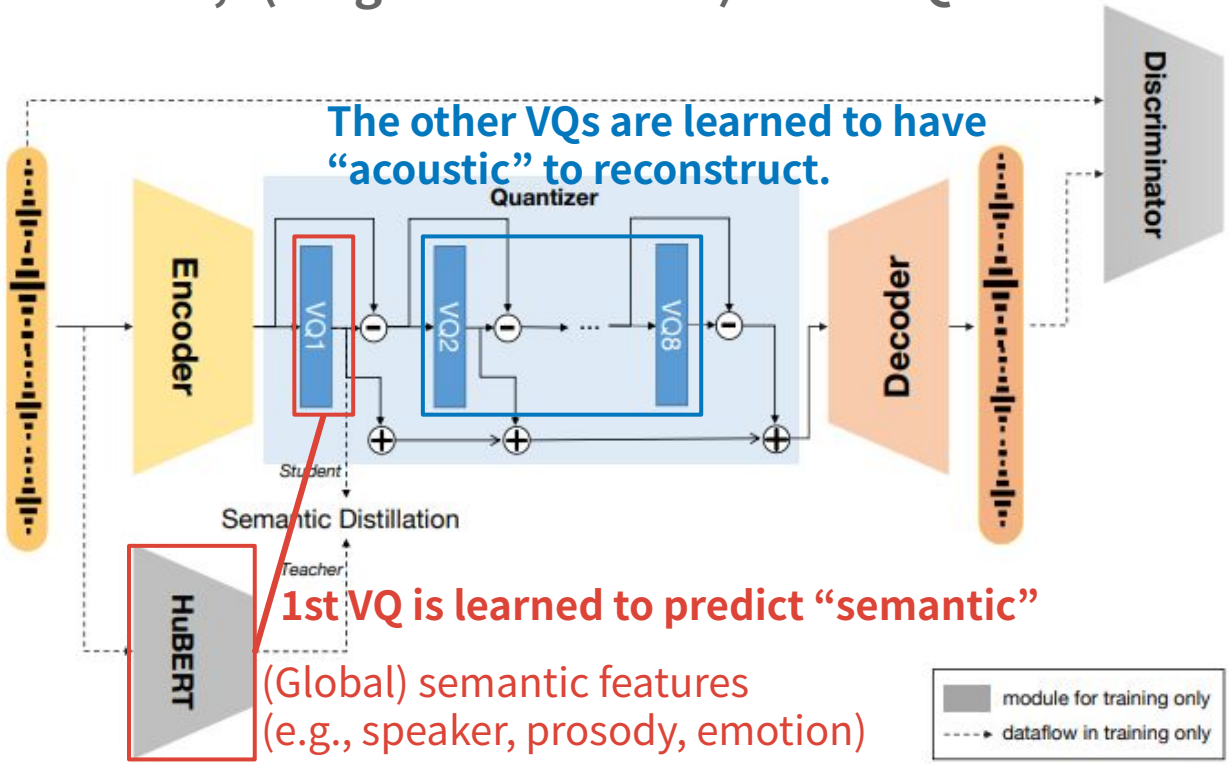
Errors from previous quantization are covered in subsequent quantization

Neural Audio Codec using RVQ



SpeechTokenizer: disentangling “semantic” and “acoustic”

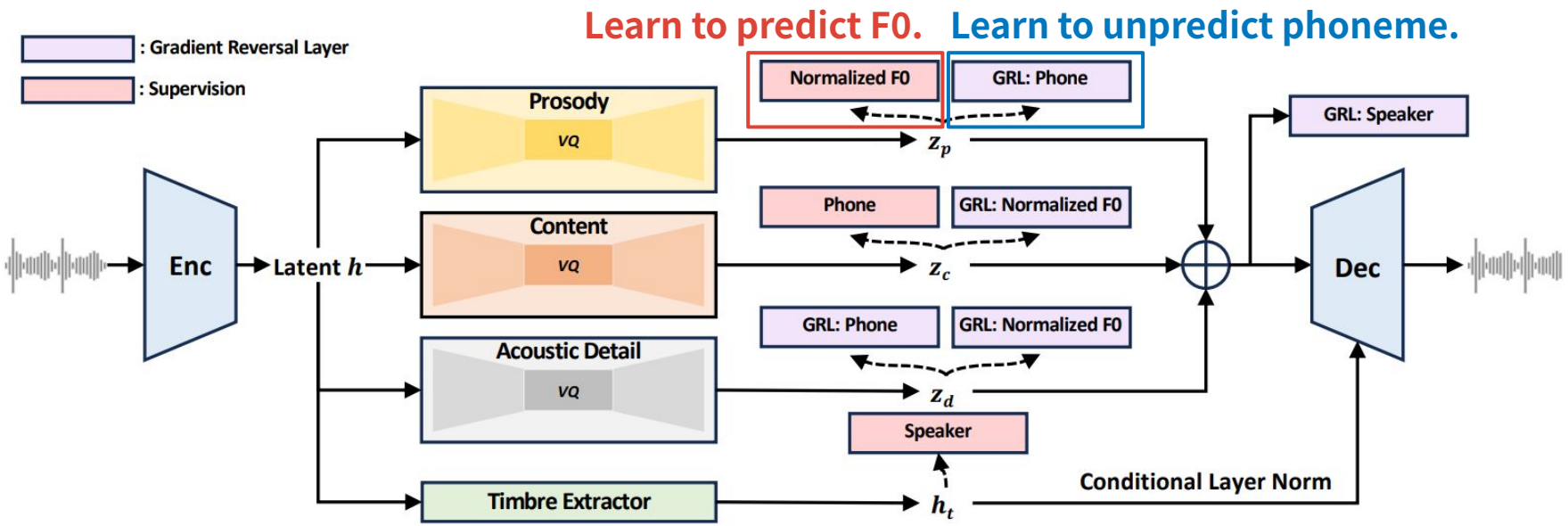
1st VQ for “semantic,” (long-term features) other VQs for “acoustic” (short)



(You will learn about HuBERT in the future class)

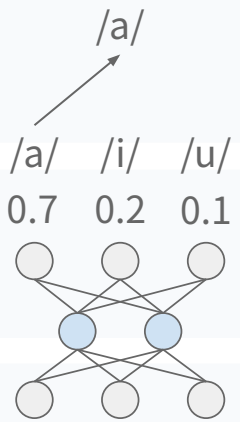
FACodec: disentangling spectrum, pitch, and speaker

Gradient reversal layers (GRLs) prevents decoded features from predicting unrelated features.



Models can process speech like language!

Speech-to-text (cont. to disr.)



Feature extraction

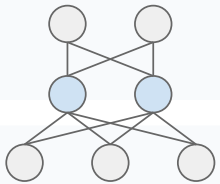
Speech

Text-to-speech (disr. to cont.)

F0 = 80.0 Hz

$N(F_0; 80, 20)$

80 20

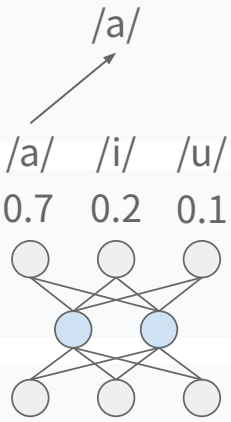


Word embedding

/a/

Text

Speech-to-text (using codecs)



Word embedding

“14” “81”...

Codec encoder

Speech

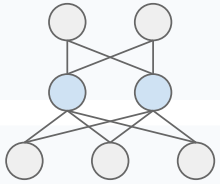
Text-to-speech (using codecs)

Codec decoder

“1”

“1” “2”

0.7 0.3



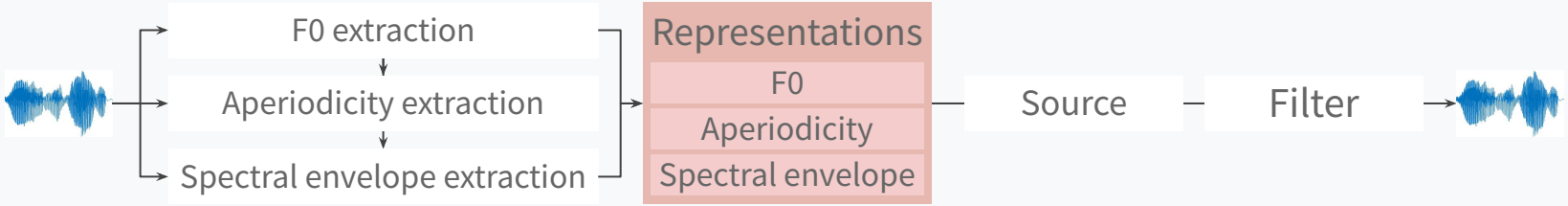
Word embedding

/a/, /i/

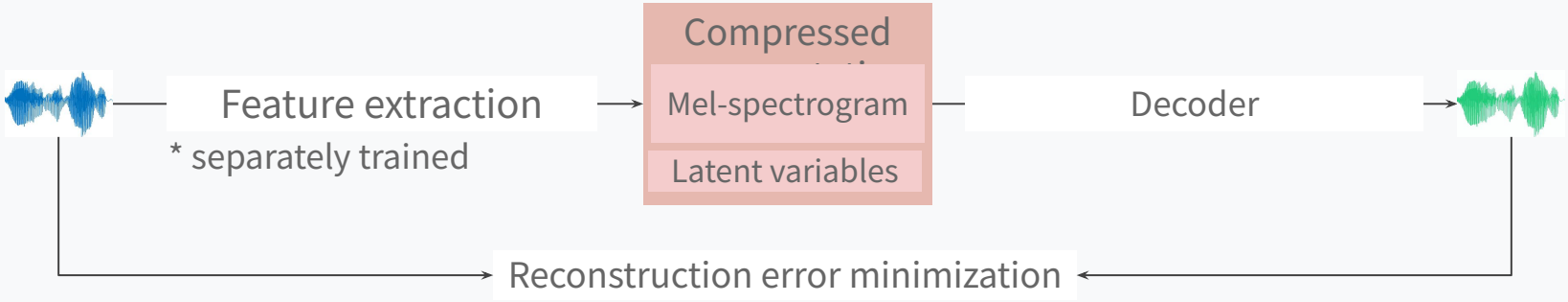
Text

From signal processing-based vocoder to neural

Signal processing-based vocoder

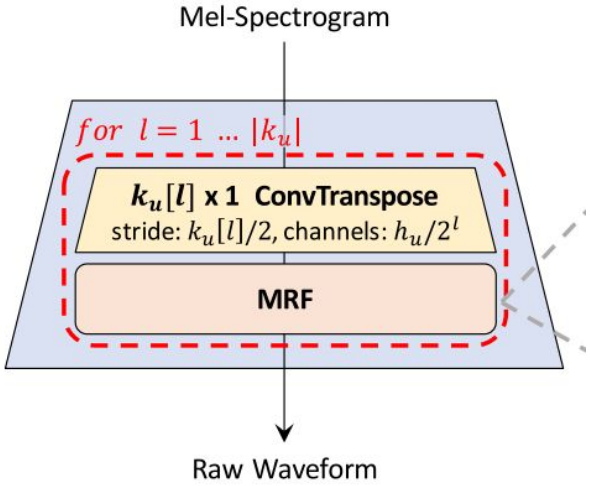


Neural vocoder

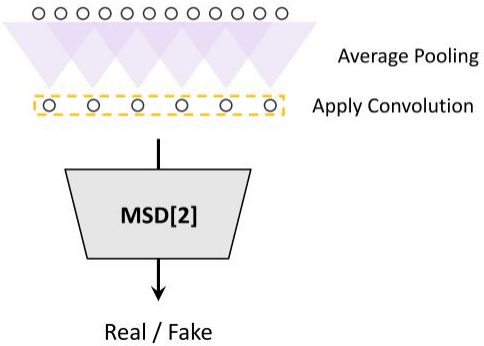


HiFi-GAN: neural vocoder standing the test of time

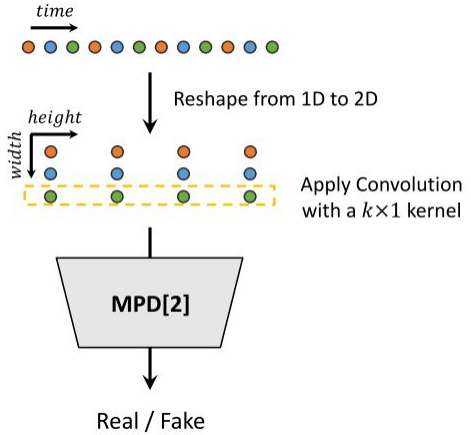
Architecture: decomvolution -based iterative upsampling



Training objective 1: multi-scale discrimination

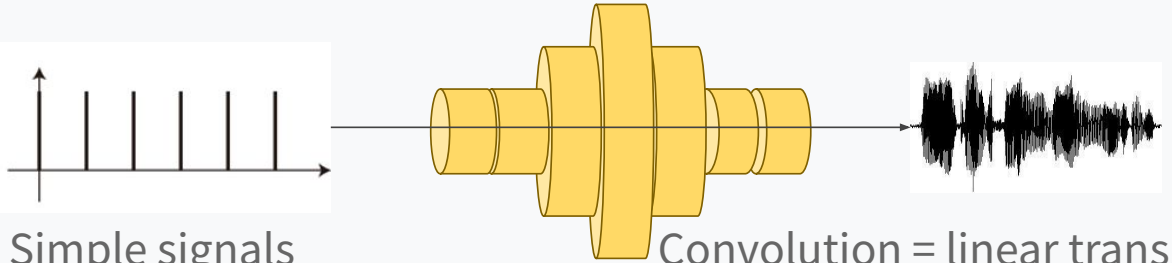


Training objective 2: multi-pitch discrimination



Neural source-filter (SiFi-GAN): disentangling source and pitch

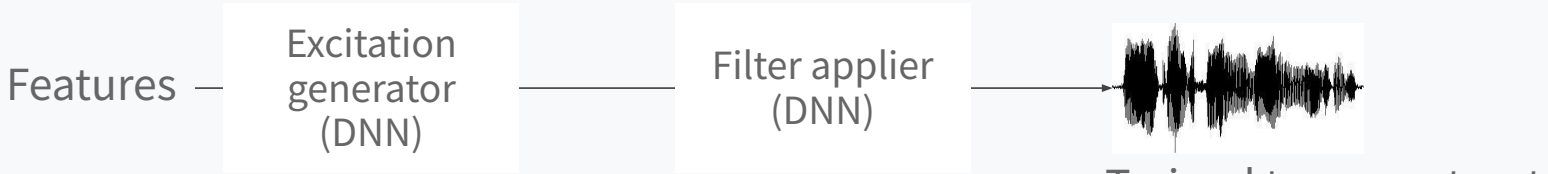
Signal processing-based vocoder (linear transformation)



Simple signals

Convolution = linear trans.

Neural vocoder (non-linear transformation)



Regularized to have characteristics of excitations.

Trained to reconstruct

Conclusion / まとめ

Conclusion

- Fundamentals of deep learning
  - Probability, linear algebra, multi-layer perceptron
  - Models to represent conditional probabilities
  - Models to transform probabilities

- Speech analysis
  - Speech features and vocoder
  - Deep learning-based analysis: speech codec
  - Neural vocoder